M. Bala Vighnesh Reddy

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EDUCATION

Indian Institute of Information Technology, Manipur **Nov 2021 – Apr 2025**
Bachelor of Technology in Computer Science and Engineering

WORK EXPERIENCE

Machine Learning Engineer **May 2026 – Present**
Aiverbalyze Technologies Private Limited

- Trained and fine-tuned a multilingual **Indic ASR system** based on the **Nemotron ASR 3.5 / FastConformer RNN-T architecture** across **12 Indic languages**, achieving an average **OIWER of 19.2%**.
- Enhanced an existing **STT inference pipeline** with **on-the-fly audio preprocessing**, **DeepFilterNet**, **Silero VAD v6**, and utterance segmentation to improve robustness for real-time voice interactions using an offline ASR model.
- Improved the existing **voice-to-voice pipeline** (STT → LLM → TTS) with **barge-in and interruption handling**, including cancellation of active **LLM generation and TTS playback** when user speech is detected.
- Generated and curated **domain-specific synthetic multilingual datasets** for ASR fine-tuning, performed **WER/error analysis**, and implemented **masked language-token conditioning** to reduce cross-language confusion, improving **Malayalam ASR performance by approximately 40%**.

Computer Vision Intern **Aug 2025 – Feb 2026**
BigThinx, Bengaluru

- Built a fashion-domain computer vision pipeline combining **U^2-Net**, **P3M-Net**, **DWPose**, and **MediaPipe** for foreground segmentation and pose estimation on 2-view image inputs.
- Tested the **SMPL parametric body model** for 3D body mesh estimation from sparse 2D observations, and used the findings to refine the pipeline direction for loose-fitting garments.
- Enhanced low-resolution and low-light inputs using **LLFormer** and **KANT**, improving the existing pipeline performance by **30%**; deployed the system on a VM with a **Gradio** interface for real-time inference.

AI/ML Intern **Jan 2025 – May 2025**
IF MedTech, Navi Mumbai

- Fine-tuned a **Vision Transformer (ViT)** on a limited medical imaging dataset and quantized it to **INT8**, achieving **89% accuracy** with reduced model size and inference time.
- Built medical image workflows using **Faster R-CNN** for anatomical region detection and **DeepLabV3** for semantic segmentation, reaching **0.64 IoU** on limited annotated data.
- Implemented a medical **RAG application** over a reference book using vector embeddings and **Llama3-8B** for context-grounded query answering.

PROJECTS

Text-to-SQL Research : **Qwen3-4B-Instruct, DeepSeek, QLoRA, GRPO, Tinker API**

- Generated and validated **9,000+ Chain-of-Thought SQL examples** for Spider and BIRD using reasoning traces from **DeepSeek**.
- Performed **supervised fine-tuning (SFT)** of **Qwen3-4B** using **4-bit QLoRA**, then applied **GRPO** with execution-based rewards to optimize SQL correctness, improving **pass@1 by 10%** over the SFT checkpoint.
- Achieved **86.7% pass@4** on Spider dev and **70.7% pass@4** on BIRD dev.

Document Querying RAG System : **LlamaIndex, Node.js, React, MongoDB** GitHub

- Developed a **Retrieval-Augmented Generation** system using **LlamaIndex** to query large document collections with context-grounded responses.
- Created **Node.js REST APIs** for document upload, indexing, and querying, backed by **MongoDB** and a React frontend with expandable chat threads.
- Implemented secure authentication and session management to protect user data.

TECH SKILLS

- **Languages & Frameworks:** Python, C/C++, TypeScript, JavaScript, FastAPI, Flask, Node.js, React.js
- **AI/ML:** PyTorch, HuggingFace, Transformers, LLMs, LoRA, QLoRA, RAG, VectorDB, LangChain, LlamaIndex, UnSloth, ASR, WER Analysis
- **Computer Vision:** ViT, Faster R-CNN, DeepLabV3, YOLOv8, EasyOCR, MediaPipe, U^2-Net, P3M-Net, segmentation, detection, pose estimation
- **MLOps & Tools:** Docker, Docker Compose, Git, GitHub, Gradio, Ollama, OpenRouter, MongoDB, MySQL, GCP

CERTIFICATIONS

- **Generative AI with Large Language Models** – DeepLearning.AI (2024)
- **Neural Networks and Deep Learning** – DeepLearning.AI / Coursera (2024)